Haiqing Gao

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Research Interests

My research focuses on mathematical modeling and stochastic optimization, with applications in supply chain management and fair resource allocation. I investigate structural properties of optimal allocation and routing policies to bridge optimization theory and practical insights for online sequential decision-making. More broadly, I develop stochastic models that provide actionable managerial insights and support interpretable decision-making.

Education

Northwestern University
Ph.D. Candidate in Industrial Engineering and Management Sciences (IEMS)
Sep. 2022 – Expected Jun. 2027

Advisor: Prof. Seyed M.R. Iravani
Ph.D. Dissertation: *Sequential Fair Allocation and Routing in Nonprofit Operations*

This dissertation develops mathematical models for fairness-aware sequential allocation and routing under uncertainty, motivated by nonprofit operations such as food banks and humanitarian relief systems. Using stochastic optimization, dynamic programming, probability, and convex analysis, it derives structural properties of optimal policies that lead to interpretable and implementable decision rules. The dissertation establishes threshold allocation policies, characterizes when demand uncertainty determines the optimal visitation order, clarifies that fairness objectives need not align with fairness metrics, and shows that optimizing routing decisions can substantially improve fairness metrics even when fairness objective value gains are modest.

The Chinese University of Hong Kong, Shenzhen
M.Phil. in Computer Information and Engineering Sep. 2020 – Jun. 2022
B.S. in Applied Mathematics (First Honor) Sep. 2016 – May 2020

Research Experience

Methodologies: Stochastic Optimization, Dynamic Programming, Probability Theory, Structural Analysis of Optimal Policies, Online Learning, Convex and Nonconvex Optimization

Applications: Nonprofit and Humanitarian Operations, Supply Chain Management, Fair Resource Allocation, Healthcare, Federated Learning

Northwestern University | Ph.D. Research
Concept Drift Detection during Online Federated Learning from Hybrid Data May 2026 – Present

Motivated by collaborative learning in privacy sensitive healthcare settings, this project investigates online federated learning under nonstationary data distributions, where hospitals seek to jointly train predictive models without sharing patients' data. We develop a federated learning framework that integrates gradient tracking, concept drift detection, and randomized client scheduling to adapt to evolving data distributions.

- Proved a dynamic regret bound for the *FedHD* gradient tracking algorithm (Gao et al., 2023) under nonstationary conditions.
- Constructed an optimistic loss estimator for detecting abrupt and gradual concept drift, compatible with Master-FL framework *(Ganguly and Aggarwal, 2024)*.
- Incorporated randomized client scheduling into federated learning and derived probabilistic convergence guarantees. *Preparing a conference manuscript.*

Sequential Fair Allocation and Collection of a Capacitated Truck with Stochastic Supply in Nonprofit Operations Apr. 2026 – Present

Motivated by food bank operations and humanitarian relief, this project studies a capacitated vehicle that sequentially visits communities with uncertain demands and donors with stochastic supplies. We formulate dynamic programs for sequential allocation, collection, and their joint optimization, where demand and supply are revealed only upon arrival, to characterize fair and implementable resource distribution policies.

- Developed dynamic programs for allocation, collection, and joint allocation and collection by a capacitated truck with stochastic supply nodes.
- Established monotonicity and threshold structures of optimal allocation and collection policies with respect to realized demand or supply and prior minimum service levels and characterized the monotonicity of the corresponding thresholds.

Multi-vehicle Sequential Fair Allocation and Routing: A Linear Approximation Model Jan. – Mar. 2026

Motivated by large-scale humanitarian distribution involving multiple vehicles, this project extends sequential fair allocation and routing from single vehicle to multiple vehicles. We formulate a stochastic mixed integer optimization model and develop a two-stage approximation that decomposes the problem into clustering and sequential allocation and routing decisions to improve computational tractability.

- Formulated a stochastic mixed integer program for multi-vehicle sequential fair allocation and routing.
- Developed a two-stage stochastic approximation that separates first-stage clustering from second-stage sequential allocation and routing.

Sequential Fair Allocation and Routing in Nonprofit Operations Jan. 2023 – Jun. 2026

Motivated by food bank operations, disaster relief, and vaccine distribution, this project studies the sequential distribution of limited resources across communities with uncertain demands. We formulate stochastic dynamic programs that jointly optimize allocation and routing under two max-min fairness objectives, *ex-post* and *forward* fairness, to investigate the structural properties of optimal policies and the implications of different fairness objectives for operational decisions.

- Established threshold structures and an equating property of optimal allocation policies under both objectives; proved the optimality of routing in decreasing order of the coefficient of variation under specified conditions for the *ex-post* objective.
- Identified differences in short- and long-term fairness metrics between *ex-post* and *forward* objectives, informing objective selection based on managerial priorities.
- Proposed the PPA-deCV heuristic for joint allocation and routing, achieving performance comparable to the optimal policy under both fairness objectives.

CUHK-Shenzhen | Master's Research
Hybrid Machine Learning via Gradient Tracking May 2020 – Jun. 2022

Motivated by collaborative machine learning with distributed and heterogeneous data, this project investigates federated learning from hybrid data, where clients possess different subsets of samples and features. We develop a gradient tracking federated learning algorithm to coordinate decentralized model training while preserving data privacy and improving convergence performance.

- Designed a federated learning algorithm using gradient tracking to learn from hybrid data and established theoretical convergence guarantees.
- Improved accuracy by 50% and achieved twice the convergence speed of the local training model.

Federated Q-learning for mHealth Interventions Oct. 2020 – Dec. 2020

Motivated by personalized mobile health interventions, this project investigates reinforcement learning across distributed patient data while preserving privacy. We develop a personalized federated Q-learning framework incorporating Bayesian learning to support adaptive treatment decisions without centralizing individual patient information.

- Developed a personalized federated Q-learning algorithm incorporating Bayesian learning.
- Achieved threefold faster convergence while maintaining accuracy through decentralized data processing.

CUHK-Shenzhen | Research Assistant Mar. 2019 – Aug. 2019
Fair Driver Order Dispatch on the Online Ride-Hailing Platform

Motivated by fairness and efficiency challenges in ride-hailing platforms, this project investigates online matching between dynamically arriving customer requests and reusable drivers under customer abandonment. We formulate a stochastic online matching problem and develop learning-based dispatch algorithms to balance matching efficiency and fairness.

- Formulated a stochastic online matching problem with customer abandonment and reusable drivers.
- Applied a multi-armed bandit approach and developed an online Hungarian bottleneck algorithm to improve matching fairness and efficiency.

Publications & Working Papers

Gao, H., S. M. R. Iravani, and S. R. Sinclair. *Sequential Fair Allocation and Routing in Nonprofit Operations.* Submitted to *Operations Research*, June 2026.

Gao, H., S. Ge, and T.-H. Chang. *FedHD: Communication-Efficient Federated Learning from Hybrid Data. Journal of the Franklin Institute*, 2023.

Gao, H., and S. M. R. Iravani. *Sequential Fair Allocation and Collection of a Capacitated Truck with Stochastic Supply in Nonprofit Operations*. Working paper.

Presentations

History Dependent or Memoryless? Objective Selection for Fair Sequential Allocation and Routing in Non-profit Operations

- 2025 INFORMS Annual Meeting — Oct. 2025
  Invited Talk for Public Sector OR (Session Chair)
- 3rd Annual Supply Chain and Operations Management Conference (SCOM) — Aug. 2025
  Invited Talk for Dedicated Student Session

Fair Allocation in Nonprofit Operations: Optimal Routing Policy

- Chicago Area Society for Industrial and Applied Mathematics (SIAM) Student Conference 2025 (CASSC25) — Mar. 2025
  Poster & Lightning Talk
- Toyota Technological Institute at Chicago (TTIC) Workshop — Aug. 2024
  Poster & Lightning Talk

Federated Q-learning in Mobile Health (mHealth) Intervention

- Reinforcement Learning Project Presentations for Graduate Students — Nov. 2020
  Poster & Lightning Talk

Fair Order Dispatch on the Online Ride-Hailing Platform

- Shenzhen Research Institute of Big Data Research Presentation — Jul. 2019

Honors

Northwestern University

Dissertation Final Year Fellowship from Transportation Center — 2026 – 2027
Institute of Industrial & Systems Engineers (IISE)
Future Faculty Fellows Program (Top 10%) — 2025 – 2026

The Chinese University of Hong Kong, Shenzhen

Master's List (Top 10%) — 2017 – 2019
Dean's List (Top 10%) — 2016 – 2019
Academic Performance Scholarship (Top 7%) — 2017

Teaching Experience

Lecturer

Northwestern University

Graduate Boot Camp Instructor for Python — Sep. 2026
Graduate Boot Camp Instructor for Probability — Sep. 2025

Teaching Assistant

Northwestern University

IEMS 365: Analytics for Social Good — Winter/Spring Quarters 2025/2026
IEMS 302: Probability — Winter/Fall Quarters 2024/2025/2026
IEMS 303: Statistics — Fall Quarter 2024
IEMS 307: Quality Improvement by Experimental Design — Spring Quarter 2024
IEMS 201: Introduction to Statistics or Data-Driven Decision-Making — Fall Quarter 2023

The Chinese University of Hong Kong, Shenzhen
MATH 3253: Complex Variables Spring Semester 2022
STA 2001: Probability and Statistics I Spring Semester 2019

Leadership & Service

The Graduate School Diversity Peer Mentor | *Northwestern University* Dec. 2023 – Apr. 2026
- Guided first-year international Ph.D. students through academic and cultural transitions.

INFORMS Peer Mentor | *Northwestern University INFORMS Student Chapter* Dec. 2023 – Aug. 2025

- Mentored two incoming IEMS Ph.D. students during their transition to graduate study.

English Language Program Foundations Peer Mentor | *Northwestern University*
Jun. – Aug. 2024 and 2025
- Led weekly adjustment meetings for 60 incoming international students and served as a mock student in teaching assessments.

McCormick Graduate Leadership Council Navigating First Year Panelist | *Northwestern University*
Oct. 2024
- Shared graduate school experiences and resources with first-year students.

INFORMS Annual Meeting Session Chair | *Public Sector OR* Apr. – Oct. 2024
- Recruited speakers and chaired a session on resource allocation in nonprofit operations.

INFORMS Student Chapter Organizer | *Northwestern University* Jan.–Mar. 2024
- Organized interdisciplinary seminars and a networking event for more than 70 students.

Skills

Programming Languages/Packages: Python, MATLAB, R, LaTeX, PyTorch, Gurobipy
Languages: Chinese (Native), English (Proficient)